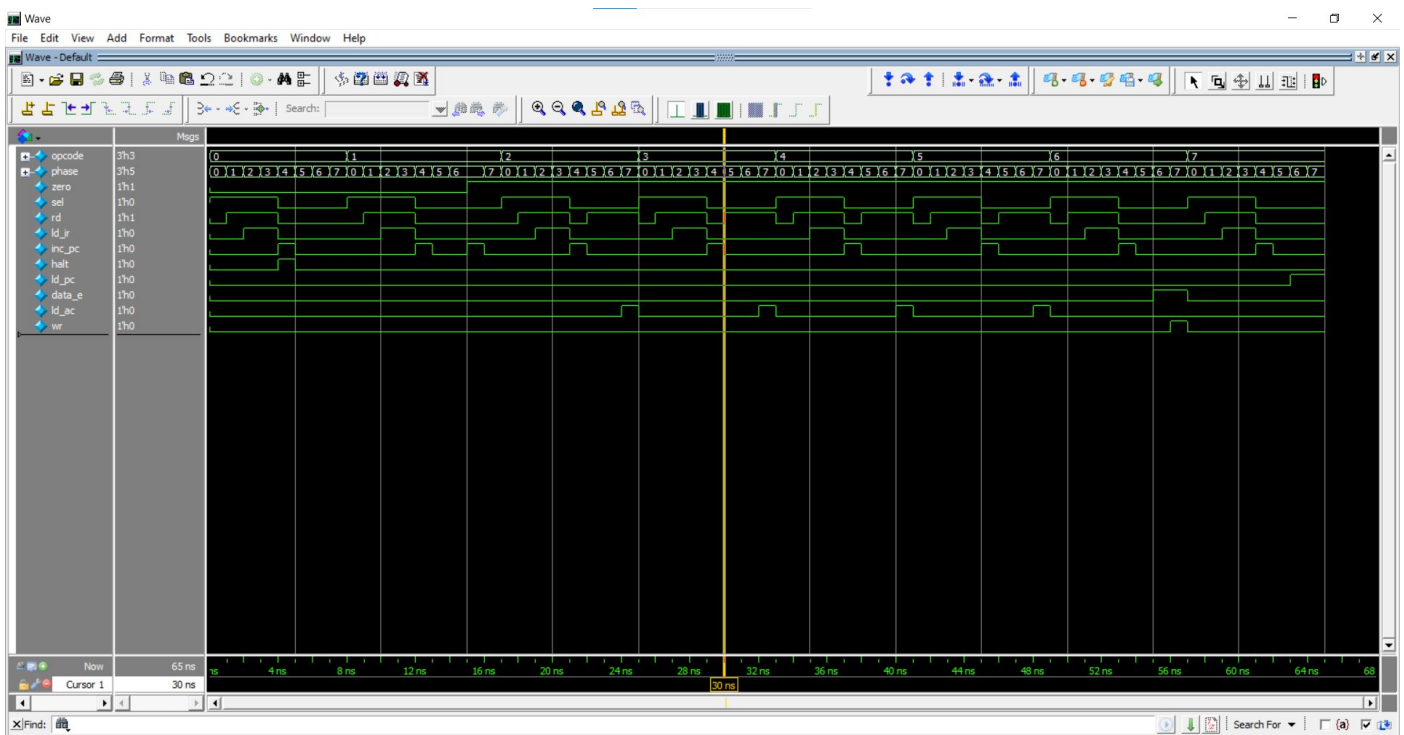


Verifying the Controller Design

TEST PASSED — every opcode across all 8 phases checked out correctly:

```
Transcript
add wave sim:/controller_test/*
VSIM 3> run -all
# Testing opcode HLT phase 0 1 2 3 4 5 6 7
# Testing opcode SKZ phase 0 1 2 3 4 5 6 7
# Testing opcode ADD phase 0 1 2 3 4 5 6 7
# Testing opcode AND phase 0 1 2 3 4 5 6 7
# Testing opcode XOR phase 0 1 2 3 4 5 6 7
# Testing opcode LDA phase 0 1 2 3 4 5 6 7
# Testing opcode STO phase 0 1 2 3 4 5 6 7
# Testing opcode JMP phase 0 1 2 3 4 5 6 7
# TEST PASSED
# ** Note: $finish      : C:/Users/hp/Desktop/Digital design assignments/VeriRISC CPU/controller_test.v(142)
#   Time: 65 ns  Iteration: 0  Instance: /controller_test
# 1
```



"The waveform above shows opcode, phase, and all control outputs sweeping through each of the 8 test opcodes (HLT through JMP) across all 8 phases. Signals such as sel, rd, and ld_ir remain high during the fetch phases (0–3) for every opcode, confirming the fetch sequence is opcode-independent. In later phases (4–7), signals like halt, inc_pc, ld_pc, wr, and data_e toggle only for the specific opcodes they correspond to (e.g. halt only for HLT, wr/data_e only for STO), matching the expected behavior in the design table."